

Section (6)

Ex(1) Drive the CDF of a distribution whose Pdf is given by

$$f(x) = \begin{cases} \frac{1}{200} (x-20) & 20 \leq x \leq 40 \\ 0 & \text{otherwise} \end{cases}$$

Also get 3 R.Os following the above distribution using the following R.N (0.49, 0.64, 0.25)

Sol $\xrightarrow{\text{cdf}}$ $F(x) = \int_{-\infty}^x f(x) dx$ $\xrightarrow{\text{pdf}}$

$$F(x) = \int_{20}^x \frac{1}{200} (x-20) dx$$

$$= \frac{1}{200} \int_{20}^x (x-20) dx$$

$$= \frac{1}{200} \cdot \frac{(x-20)^2}{2} \Big|_{20}^x$$

$$= \frac{1}{200} \left[\frac{(x-20)^2}{2} - 0 \right]$$

$$F(x) = \frac{(x-20)^2}{400}$$

↓ To get R.Os

$$R = \frac{(x-20)^2}{400}$$

$$0 \leq R \leq 1$$

$$20 \leq x \leq 40$$

$$(x-20)^2 = 400R$$

$$x-20 = 20\sqrt{R}$$

$$x = 20\sqrt{R} + 20$$

$$x = 20(\sqrt{R} + 1)$$

$$\Rightarrow R.N = 0.49 \quad x = 20(\sqrt{0.49} + 1) = 34$$

$$\Rightarrow R.N = 0.64 \quad x = 20(\sqrt{0.64} + 1) = 36$$

$$\Rightarrow R.N = 0.25 \quad x = 20(\sqrt{0.25} + 1) = 30$$

x must be in Range $20 \leq x \leq 40$

Ex(2) Determine mean of The Random variable whose cdf is given by;

$$F(x) = x^2 \quad 0 \leq x \leq 1$$

use The following 6 Random numbers To get The Simulated mean and calculate The error in The mean. Prove That The increase in Sample Size will decrease The error.

R.N: 0.628, 0.207, 0.662, 0.774, 0.360, 0.951

Sol

$$F(x) = x^2$$

$$f(x) = \frac{d}{dx} F(x)$$

$$f(x) = 2x$$

$$\text{mean} = \int_{-\infty}^{\infty} x f(x) dx = \int_0^1 x \cdot 2x dx = 2 \int_0^1 x^2 dx = 2 \cdot \frac{x^3}{3} \Big|_0^1 = \frac{2}{3}$$

$$\text{mean} = \frac{2}{3} \Rightarrow \text{Theoretical mean}$$

To get Simulated mean

$$R = F(x)$$

$$R = x^2$$

$$x = \sqrt{R}$$

$$0 \leq R \leq 1$$

$$0 \leq x \leq 1$$

R	$x = \sqrt{R}$
0.628	0.792
0.207	0.454
0.662	0.814
0.774	0.879
0.360	0.6
0.951	0.975

$$\text{Sim}(\text{mean}) = \frac{\sum_{i=1}^6 x_i}{6}$$

$$\text{Sim}(\text{mean}) = \frac{0.792 + 0.454 + 0.814 + 0.879 + 0.6 + 0.975}{6}$$

$$\therefore \text{Sim}(\text{mean}) = 0.7523$$

$$E(\text{mean}) = \left| \frac{\text{Th}(\text{mean}) - \text{Sim}(\text{mean})}{\text{Th}(\text{mean})} \right| \times 100$$

$$E_i(\text{mean}) = \left| \frac{\frac{2}{3} - 0.7523}{\frac{2}{3}} \right| \times 100$$

$$= 12.8 \%$$

R	$R_2 = 1 - R$	$x = \sqrt{R_2}$
0.628	0.372	0.609
0.207	0.793	0.891
0.662	0.338	0.581
0.774	0.226	0.475
0.360	0.640	0.8
0.951	0.049	0.221

$$\text{Sim}(\text{mean}) = \frac{\sum_{i=1}^{12} x_i}{12}$$

$$\text{Sim}(\text{mean}) = \frac{0.792 + 0.454 + 0.814 + 0.879 + 0.6 + 0.975 + 0.609 + 0.891 + 0.581 + 0.475 + 0.8 + 0.221}{12}$$

$$\text{Sim}(\text{mean}) = 0.67425$$

$$E_2(\text{mean}) = \left| \frac{\frac{2}{3} - 0.67425}{\frac{2}{3}} \right| \times 100 = 1.14\%$$

$$E_1(\text{mean}) > E_2(\text{mean})$$

Then increasing in sample size will decrease the Error

Ex(3)

$$f(x) = \begin{cases} 1+x & -1 \leq x \leq 0 \\ 1-x & 0 \leq x \leq 1 \end{cases}$$

R.N: 0.73, 0.27, 0.5, 0.81, 0.33

Sol

$$F_1(x) = \int_{-1}^x (1+x) dx$$

$$F_1(x) = \frac{(1+x)^2}{2} \Big|_{-1}^x$$

$$F_1(x) = \frac{(1+x)^2}{2}$$

$$R = \frac{(1+x)^2}{2}$$

$$(1+x)^2 = 2R$$

$$1+x = \sqrt{2R}$$

$$x_1 = \sqrt{2R} - 1$$

$$0 \leq R \leq 0.5$$

$$-1 \leq x \leq 0$$

$$F_2(x) = \int_{-1}^0 (1+x) dx + \int_0^x (1-x) dx$$

$$F_2(x) = \frac{(1+x)^2}{2} \Big|_{-1}^0 + \left(-\frac{(1-x)^2}{2} \Big|_0^x \right)$$

$$F_2(x) = \frac{1}{2} - \frac{(1-x)^2}{2} + \frac{1}{2}$$

$$F_2(x) = 1 - \frac{(1-x)^2}{2}$$

$$R = 1 - \frac{(1-x)^2}{2}$$

$$\frac{(1-x)^2}{2} = 1 - R$$

$$1-x = \sqrt{2(1-R)}$$

$$x_2 = 1 - \sqrt{2(1-R)}$$

$$0.5 \leq R \leq 1$$

$$0 \leq x \leq 1$$

R.N 0.73

$$x_2 = 1 - \sqrt{2(1-0.73)} = 0.265$$

R.N 0.27

$$x_1 = \sqrt{2 \times 0.27} - 1 = -0.265$$

R.N 0.5

$$x_1 = \sqrt{2 \times 0.5} - 1 = 0$$

$$\text{or } x_2 = 1 - \sqrt{2(1-0.5)} = 0$$

R.N 0.81

$$x_2 = 1 - \sqrt{2(1-0.81)} = 0.383$$

R.N 0.33

$$x_1 = \sqrt{2 \times 0.33} - 1 = -0.187$$